

Quantum 20 Minute Discovery

Quantum hackathon brief · MSc cohort · Groups of approx. 5

The Exercise

In groups of roughly five, discover and explain one thing about quantum that none of you knew before today and that is genuinely new or surprising to you.

It can be in blockchain, but also physics, computing, communication, cryptography, or its philosophical implications.

Deliver a 60-second explanation with a single slide or sketch or whiteboard panel.

Pedagogical Framing

This is developing a hard won skill: locating accurate, well-supported, and non obvious information and explaining it cleanly. Use AI as a research partner, but not just as a full writer.

What is Interesting / Surprising

- It's not common knowledge or obvious to your peers (superposition, Schrödinger's cat etc.)
- It is true and well-supported (academic / research supported)
- You can explain the idea succinctly

How to Works

Pick someone to pitch. You all research. You may use ai. You should ask it for sources. You should ask questions amongst yourselves. What is the most important or interesting thing you would like to hear about?

Cite and follow up sources. Could you go away and research this further based on your work? Can you convince the rest of the class that this is interesting?

Main Items

1. Is it surprising / interesting?
2. Is it well explained? In your own voice (not ai)?
3. Is it well supported?

Calibration, a strong pitch looks like (but not one of these)

- Superdense coding sends two classical bits using one qubit plus pre-shared entanglement.
- Quantum gravimeters can map underground voids — pipes, tunnels, archaeology — through metres of rock.
- The delayed-choice quantum eraser appears to let a future measurement change a past outcome — and why that intuition is wrong.
- "Quantum supremacy" results typically solve problems engineered to be hard classically, not useful ones.
- The no-cloning theorem is the reason quantum money could, in principle, be unforgeable.

Prompting

Prompt, read, think, verify, follow-links, read, prompt.